

FN585 Financial Modelling & Dealing

Week 5

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Estimating the lag length using information criteria

Lag length selection for AR

- Including more lags will always improve fit (R^2), but estimating too many parameters increases estimation error.
- One could take F-statistic approach of starting with a large lag, say AR(6) and testing whether the coefficient of the 6th lag is significant at 5 percent level: if not drop it and estimate AR(5), etc.
- However, even when the true model is AR(5), this method would end up with AR(6) 5 percent of times.

Estimating the lag length using information criteria

Bayes information criteria (BIC)

- Estimate lag p by minimizing:

$$BIC(p) = \ln \left[\frac{RSS(p)}{T} \right] + (p + 1) \frac{\ln(T)}{T}$$

- When you increase p , you get better fit so lower $SSR(p)$, but the second term above is a penalty term

Estimating the lag length using information criteria

Akaike information criteria (AIC)

- Estimate lag p by minimizing:

$$AIC(p) = \ln \left[\frac{RSS(p)}{T} \right] + (p + 1) \frac{2}{T}$$

- Whether you use BIC or AIC between competing AR lags, all regressions must be estimates using the same observations.

Forecasting uncertainty

- Forecasts are inherently uncertain and error ridden
- Good model selection minimizes forecast errors, but does not eliminate them: report forecast uncertainty
- In practice, summary measures of forecast uncertainty are based on the Mean Squared Forecasting Error (MSFE)

$$MSFE = E[(Y_{T+1} - \hat{Y}_{T+1|T})^2]$$

- The MSFE is unknown and must be estimated.

Use of the MSFE

- Three ways of estimating the MSFE (next week, we focus on R-implementation)
- An estimate of the MSFE should be reported alongside the forecast: A forecast interval gives a range of forecasts
- A forecast interval is a good way of conveying forecast uncertainty: wider intervals indicate greater uncertainty

Estimating MSFE

- We will focus on AR(p) model for illustration.
- MSFE has two components, randomness of the future value and randomness due to estimation of model.

$$MSFE = \sigma_u^2 + var(\hat{\beta}_0 + \hat{\beta}_1 Y_T + \cdots + \hat{\beta}_p Y_{T-p+1})$$

Method 1: standard error of regression (SER)

- The second term is proportional to $\frac{1}{T}$. So when T is very large, the first term dominates.

Method 2: final prediction error

- When T is not too large, if we assume homoscedasticity of errors, one has

$$\text{var}(\hat{\beta}_0 + \hat{\beta}_1 Y_T + \cdots + \hat{\beta}_p Y_{T-p+1}) \sim \hat{\sigma}_u^2[(p+1)/T]$$

and

$$\widehat{MSFE}_{FPE} = \frac{T+p+1}{T} \hat{\sigma}_u^2 = \frac{T+p+1}{T-p-1} \frac{RSS}{T}$$

$$\widehat{MSFE}_{SER} = \hat{\sigma}_u^2 = \frac{RSS}{T-p-1}$$

Estimating MSFE

Method 3: pseudo out of sample forecasting

- Training data: first $(T - p)$ data points
- Reserved sample: final p observations

Steps

1. Choose p . Denote $s = T - p$
2. Estimate the forecasting regression using observations $t = 1, \dots, s$
3. Compute forecast for $s+1$, call this $\hat{Y}_{s+1|s}$
4. Compute forecast error $\tilde{u}_{s+1} = Y_{s+1} - \hat{Y}_{s+1|s}$
5. Repeat steps 2-4 for the remaining periods (reestimate the model for each period) $s = T - p - 1$ to $s = T - 1$

$$\widehat{MSFE}_{POO} = \frac{1}{p} \sum_{s=T-p+1}^T \tilde{u}_s^2$$

Forecast intervals

- A 95% forecast interval contains the future value of the forecasted variable in 95% of repeated applications
- A forecast interval is like a confidence interval, but refers to the value of the forecasted variable (not a population coefficient)
- Unlike confidence intervals, however, forecast intervals are typically based on an assumption of normal errors (Distribution of OLS estimator $\hat{\beta}$ is well approximated by normal distribution in large samples, even when errors are non-normal; that is not the case for a forecast Y_{T+1})

Forecast intervals

$AR(p)$ model for Y

- Stationary $Y \sim AR(p)$ model with i.i.d normal errors:

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots + \beta_p Y_{t-p} + u_t$$

where $u_t \sim N(0, \sigma_u^2)$.

- Using $\{Y_t, t = 1, 2, \dots, T\}$ estimate the model by OLS
- The forecast error

$$Y_{T+1} - \hat{Y}_{T+1|T} =$$

$$u_{T+1} - [(\hat{\beta}_0 - \beta_0) + (\hat{\beta}_1 - \beta_1)Y_T + \cdots + (\hat{\beta}_p - \beta_p)Y_{T-p+1}]$$

follows a 0-mean normal distribution with variance MSFE

95% forecast interval

- Since $Y_{T+1} - \hat{Y}_{T+1|T} \sim N(0, MSFE)$, it is the case that

$$P[|Y_{T+1} - \hat{Y}_{T+1|T}| < 1.96\sqrt{MSFE}] = 0.95$$

- It follows that the 95% forecast interval for Y_{T+1} is

$$[\hat{Y}_{T+1|T} - 1.96\sqrt{\widehat{MSFE}}, \hat{Y}_{T+1|T} + 1.96\sqrt{\widehat{MSFE}}]$$

where $\widehat{MSFE}$ estimates $MSFE$

- A 95% forecast interval contains Y_{T+1} in 95% of repeated applications: wider intervals indicate greater uncertainty

Estimating MSFE

We will focus on AR(p) model for illustration.

MSFE has two components, randomness of the future value and randomness due to estimation of model.

$$MSFE = \sigma_u^2 + var(\hat{\beta}_0 + \hat{\beta}_1 Y_T + \cdots + \hat{\beta}_p Y_{T-p+1})$$

Method 1: Standard Error of Regression (SER)

The second term is proportional to $1/T$. So when T is **very large**, the first term dominates.

$$\widehat{MSFE}_{SER} = \hat{\sigma}_u^2 = \frac{RSS}{T - p - 1}$$

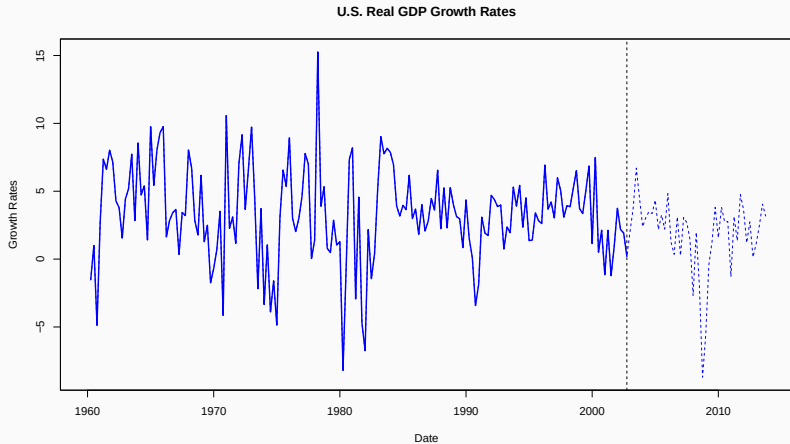
Forecast Intervals for US GDP Growth

```
# load US macroeconomic data
usmac_qt <- read_xlsx("us_macro_quarterly.xlsx", sheet = 1, col_types = c("text", rep("numeric", 9)))

## New names:
## * ' ' -> '...1'

# Fix format of date
usmac_qt$...1 <- as.yearqtr(usmac_qt$...1, format = "%Y:O%q")
# Relabel column names in dataframe
colnames(usmac_qt) <- c("Date", "GDPC96", "JAPAN_IP", "PCEPTI",
                        "GS10", "GS1", "TB3MS", "UNRATE", "EXUSUK", "CPIAUCSL")
# GDP series as xts object
GDP <- xts(usmac_qt$GDPC96, usmac_qt$Date)["1960::2013"]
# GDP growth series as xts object
GDPGrowth <- xts(400 * log(GDP/lag(GDP)))
# Training data
GDPGrowth.train <- na.omit(GDPGrowth["1960::2002"])
```


US GDP GROWTH



An AR(2)-Model for US GDP Growth

- Suppose GDP growth follows an AR(2)-model with iid errors:

$$GDPR_t = \beta_0 + \beta_1 GDPR_{t-1} + \beta_2 GDPR_{t-2} + u_t$$

- where $E[u_t | GDPR_{t-1}, GDPR_{t-2}, \dots] = 0$.
- Estimate model by OLS using $GDPR_t$; $t = 1, 2, \dots, T$
- Suppose further that $u_t \sim N(0, \sigma_u^2)$, so the 95% forecast interval for $GDPR_{t+1}$ is

$$[\widehat{GDPR}_{t+1|T} - 1.96\sqrt{\widehat{MSFE}}, \widehat{GDPR}_{t+1|T} + 1.96\sqrt{\widehat{MSFE}}]$$

Estimate AR(2) model for US GDP growth

```
##
## Time series regression with "ts" data:
## Start = 3, End = 171
##
## Call:
## dynlm(formula = ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)) +
##       L(ts(GDPGrowth.train), 2))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6623  -1.7626  -0.0594   1.9502  12.9331
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.97146    0.39513   4.989 1.52e-06 ***
## L(ts(GDPGrowth.train))  0.24888    0.07665   3.247 0.00141 **
## L(ts(GDPGrowth.train), 2) 0.16723    0.07624   2.193 0.02967 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.275 on 166 degrees of freedom
## Multiple R-squared:  0.1152, Adjusted R-squared:  0.1045
## F-statistic: 10.8 on 2 and 166 DF, p-value: 3.881e-05
```

One-step ahead forecast of GDP growth

```
T <-length(GDPGrowth.train) # Length of observed series
fcast.ar2 <- ar2$coefficients %*% c(1, GDPGrowth.train[T], GDPGrowth.train[(T-1)])
fcast.ar2 # Print to console
```

```
##           [,1]
## [1,] 2.340301
```

95% forecast interval for US GDP growth

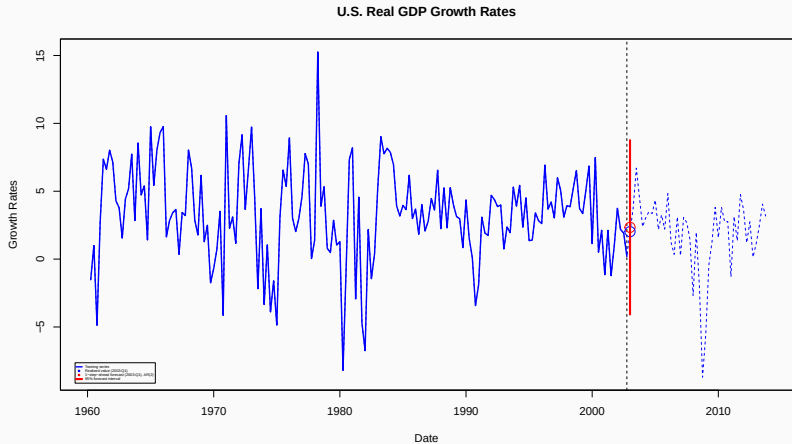
- Since $T \gg p$, use $\widehat{MSFE} = SER = s_u^2$

```
lb <- fcast.ar2 - 1.96*summary(ar2)$sigma
ub <- fcast.ar2 + 1.96*summary(ar2)$sigma
c(lb, ub)
```

```
## [1] -4.078784 8.759385
```

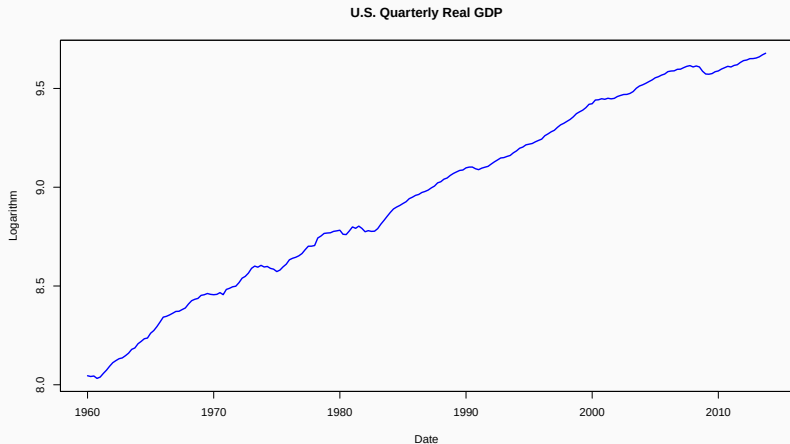
- Hence, 95% forecast interval from our AR(2) -model for US GDP growth is $[-4.08, 8.76]$, rather wide!

US GDP Growth, Forecast, and 95% Forecast Interval



Trends

A trend is a persistent long-term movement in a variable over time; a time series fluctuates around its trend.



Deterministic and Stochastic Trends

- A deterministic trend is a nonrandom function of time t ; a linear trend is a deterministic trend:

$$Y_t = \alpha + \delta t + \beta_1 Y_{t-1} + u_t$$

- Deterministic trends typically add no complications to inference and forecasting.
- Stochastic trends are random and vary over time: may be growing for a while, and then start falling.
- Stochastic trends complicate inference and forecasting, but often more appropriate for economic time series.

Random Walk

The random walk model is a simple and versatile way of introducing a stochastic trend to a time series model.

- Y follows a random walk if changes in Y are i.i.d.:

$$\Delta Y_t = u_t \iff Y_t = Y_{t-1} + u_t,$$

where u_t is i.i.d. and $E[u_t | Y_{t-1}, Y_{t-2}, \dots] = 0$.

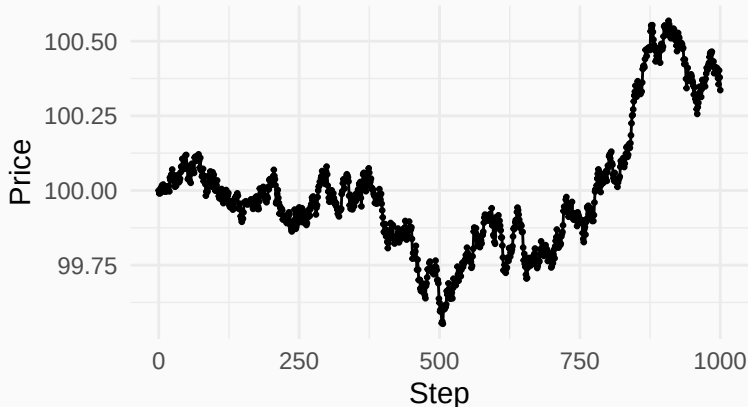
- Y_t is produced by a “random step” u_t from last period’s position Y_{t-1} ; hence, the “random walk” label!
- A random walk is nonstationary because the variance increases over time.

A Simulated Random Walk

The model for the random walk is defined as:

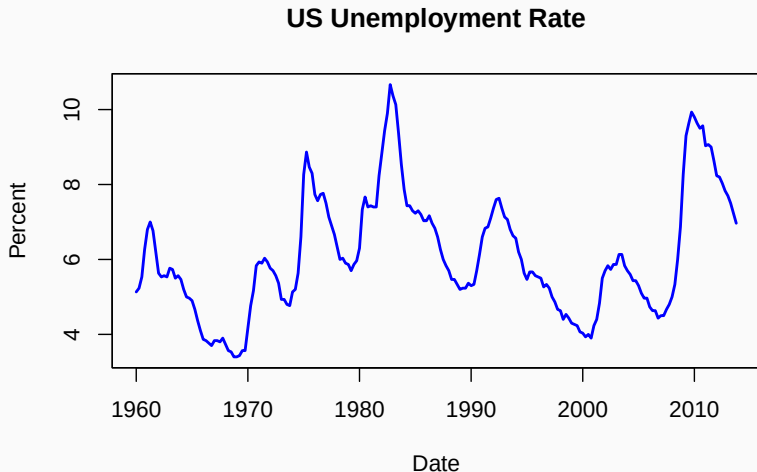
$$Y_t = Y_{t-1} + u_t; \quad u_t \sim \mathcal{N}(0, 1); \quad Y_0 = 0$$

Random Walk with Normal Distribution



A “Real Life” Random Walk?

Observing the US Unemployment Rate, we might consider if it follows a random walk pattern.



Random Walk with a Drift

- An persistent upward (or downward) tendency is hard to capture in the pure random walk.
- Add a drift β_0 to the pure random walk model:

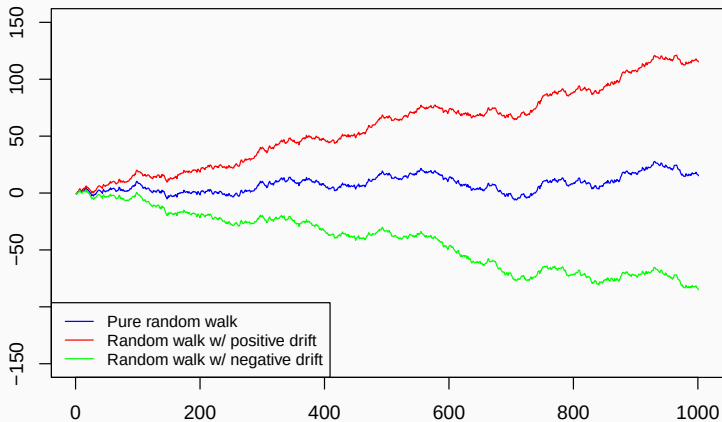
$$\Delta Y_t = \beta_0 + u_t \iff Y_t = \beta_0 + Y_{t-1} + u_t,$$

where u_t is i.i.d. and $E[u_t | Y_{t-1}, Y_{t-2}, \dots] = 0$.

- Y_t is produced by a “random step” $\beta_0 + u_t$ from last period’s position Y_{t-1} .
- $\beta_0 > 0$: upwards drift; $\beta_0 < 0$: downwards drift.

Simulated Random Walks with Drifts

$$Y_t = \beta_0 + Y_{t-1} + u_t; \quad u_t \sim \mathcal{N}(0, 1); \quad Y_0 = 0$$



Unit Roots

- The $AR(1)$ model is $Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$; hence, a random walk is an $AR(1)$ model with $\beta = 1$.
- An $AR(1)$ model with $\beta = 1$ is said to have a unit root (because the so-called characteristic polynomial of the $AR(1)$ model, $1 - \beta_1 z$, in that case has root $z = 1$).
- Unit root process, random walk process, and stochastic trend all refer to $Y_t = \beta_0 + Y_{t-1} + u_t$ with u_t i.i.d.

Non-Normal Sampling Distributions

Consider the $AR(1)$ model

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

with u_t i.i.d. and $E[u_t | Y_{t-1}, Y_{t-2}, \dots] = 0$.

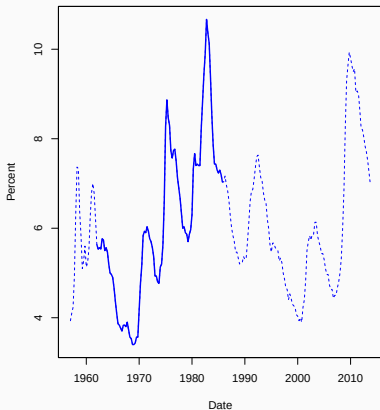
- If it is in fact the case that $\beta_1 = 1$, then
 - $\hat{\beta}_1$ severely biased towards 0 for small T
 - Distribution of $\hat{\beta}_1$ not normal, even for large T
- Problematic for constructing forecasts, forecast intervals, and for testing hypotheses involving β_1

Spurious Regression

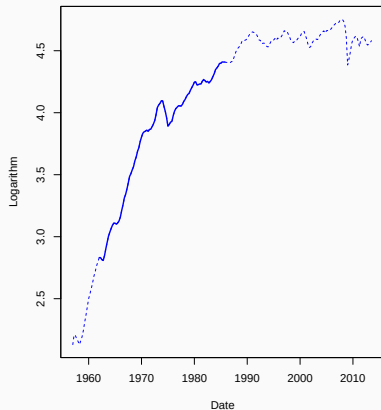
- Two nonstationary time series may appear related, when in fact they are not! This is called spurious regression.
- If used in a forecast context, spurious regressions lead to biased forecasts.

JPN INDUSTRIAL PRODUCTION AND US UNEMPLOYMENT 1962-1985

US Unemployment Rate



Log of the JPN Industrial Production Index

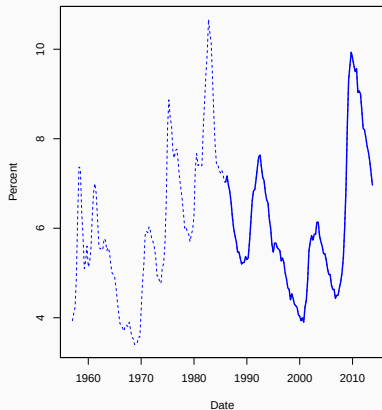


JPN INDUSTRIAL PRODUCTION AND US UNEMPLOYMENT 1962-1985

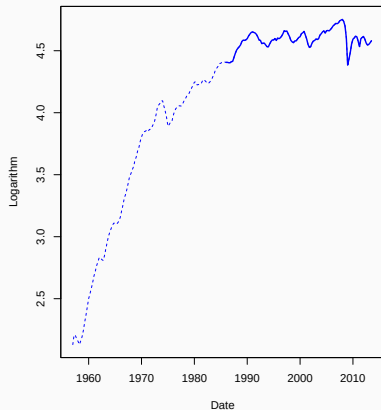
```
##
## =====
##                               Dependent variable:
##                               -----
##                               us.u.62.85
## -----
## jpn.logindu.62.85            2.221***
##                               (0.311)
##
## Constant                     -2.375*
##                               (1.196)
##
## -----
## Observations                 96
## R2                           0.352
## Adjusted R2                  0.345
## Residual Std. Error          1.433 (df = 94)
## F Statistic                   51.003*** (df = 1; 94)
## =====
## Note:                        *p<0.1; **p<0.05; ***p<0.01
```

JPN INDUSTRIAL PRODUCTION AND US UNEMPLOYMENT 1986-2013

US Unemployment Rate



Log of the JPN Industrial Production Index



JPN INDUSTRIAL PRODUCTION AND US UNEMPLOYMENT 1986-2013

```
##
## =====
##                      Dependent variable:
##                      -----
##                      us.u.86.13
## -----
## jpn.logindu.86.13      -7.932***
##                      (1.743)
##
## Constant              42.518***
##                      (8.003)
##
## -----
## Observations           111
## R2                     0.160
## Adjusted R2            0.152
## Residual Std. Error    1.400 (df = 109)
## F Statistic            20.714*** (df = 1; 109)
## =====
## Note:                  *p<0.1; **p<0.05; ***p<0.01
```

Order of Integration

- If Y_t follows a random walk, then $\beta_1 - 1 = 0$ and the first difference of Y_t becomes $\Delta Y_t = v_t$.
- Time Series like Y_t , which can be made stationary by taking the first difference, are said to be integrated of order one, denoted as $I(1)$. Stationary series are said to be **integrated** of order zero, $I(0)$.
- In general, the order of integration of a series is the minimum number of times it must be differenced to make it stationary.

Common Stochastic Trends and Cointegration

- The special case where two series share the *same stochastic trend* is interesting.
- If two series have a common stochastic trend, we say the series are cointegrated.
- For **cointegrated** series, it is possible to run reliable, i.e., nonspurious, regressions.

How to Deal with Nonstationary Time Series

- If a series looks nonstationary, worry about the presence of stochastic trends.
- If the stochastic trend is a random walk, the first differenced series will be stationary:

$$Y_t = Y_{t-1} + u_t \iff \Delta Y_t = u_t$$

- There are formal statistical tests for nonstationary (Dickey-Fuller tests)

How to Deal with Nonstationary Time Series

- How we convert non-stationary Time Series to stationary Time Series, and the type of model we estimate, depend on whether the variables are difference stationary or trend stationary.
- In the former case, we transform the non-stationary series to stationary by taking first differences.
- In the latter case, we transform the non-stationary series to stationary by de-trending.

Unit Root Tests for Stationarity

- How do we detect whether a Time Series process is (non-)stationary?
- The most popular tool is the **Dickey Fuller test**.
- It has many variants depending on the type of the trend (r.w., r.w. with drift, with time trend, etc.).

Dickey Fuller Test with No Constant and No Trend

- The AR(1) process $Y_t = \beta_1 Y_{t-1} + v_t$ is stationary when $|\beta_1| < 1$. But, when $\beta_1 = 1$, it becomes the non-stationary r.w. process we saw in the last lecture.
- We wish to test whether β_1 is equal to one or significantly less than one.
- Tests for this purpose are known as unit root tests for stationarity.

Dickey Fuller Test with No Constant and No Trend

- A more convenient form for the test is:

$$Y_t - Y_{t-1} = \beta_1 Y_{t-1} - Y_{t-1} + v_t \iff \Delta Y_t = (\beta_1 - 1)Y_{t-1} + v_t$$

- The null hypothesis can then be written as:
 - $H_0 : (\beta_1 - 1) = 0$ against the alternative
 - $H_1 : (\beta_1 - 1) < 0$.

Dickey Fuller Test with Constant and No Time Trend

- The second Dickey Fuller test includes a constant term in the test equation:

$$\Delta Y_t = \mu + (\beta_1 - 1)Y_{t-1} + v_t$$

- The null hypothesis is unchanged: $H_0 : (\beta_1 - 1) = 0$ against the alternative $H_1 : (\beta_1 - 1) < 0$.

Dickey Fuller Test with Constant and with Time Trend

- The third Dickey Fuller test includes a constant term in the test equation and a time trend:

$$\Delta Y_t = \mu + (\beta_1 - 1)Y_{t-1} + \lambda t + v_t$$

- The null hypothesis again is unchanged:
 - $H_0 : (\beta_1 - 1) = 0$ against the alternative
 - $H_1 : (\beta_1 - 1) < 0$.

Dickey Fuller Test

- For all three variants of the D.F. test, if we reject the null hypothesis, we conclude the Time Series is stationary.
- Under $H_0 : \beta_1 = 1$
- To test the H_0 in all three variants, we estimate the test equation by OLS and examine the t-statistic for the hypothesis that $\beta_1 - 1 = 0$.

Dickey Fuller Critical Values

- The t-statistic no longer follows the t-distribution. Instead, we use the statistic often called a Dickey-Fuller t-statistic, or τ (tau) statistic.
- To carry out a one-tail test of significance, if the critical value of the Dickey-Fuller t-statistic is τ_c , we reject the H_0 if $\tau \leq \tau_c$. If $\tau > \tau_c$, then we do not reject the H_0 that the series is non-stationary.

Dickey Fuller Test Asymptotic Critical Values

The following table presents the critical values for the Dickey Fuller test at different significance levels:

Condition	10%	5%	1%
no constant, no trend	-1.62	-1.94	-2.56
constant only	-2.57	-2.86	-3.43
constant and trend	-3.12	-3.41	-3.96

Dickey Fuller Testing Procedures

First plot the time series of the variable and select a suitable Dickey-Fuller test based on a visual inspection of the plot. If the series appears to:

- be wandering or fluctuating around a sample average of zero, use the first variant of the test.
- be wandering or fluctuating around a sample average which is nonzero, use the second variant of the test.
- be wandering or fluctuating around a linear trend, use the third variant of the test.